Sonal Kamble

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EDUCATION

State University of New York - New Paltz — New Paltz, New York
MS in Computer Science — *January 2025 – Present*

Sunbeam Institute of Technology — Pune, India
PG Diploma in Advanced Computing — *August 2021 – April 2022*

DY Patil's Ramrao Adik Institute of Technology — Navi Mumbai, India
Bachelor's of Engineering in Electronics and Telecom. — *Aug. 2017 – May 2021*

EXPERIENCE

Teacher of Record – CS3 — August 2025 – Present
SUNY New Paltz — *New Paltz, NY*

- Assist professor in delivering lectures and grading assignments for **advanced programming concepts** including **Java, data structures, and algorithms**, supporting a class of 50+ students.
- Hold weekly **office hours** to provide one-on-one academic support, clarify complex topics, and mentor students in best coding practices.
- Facilitate **lab sessions**, guiding students through programming exercises, **debugging techniques**, and algorithmic problem-solving.
- Collaborate with faculty to design **instructional materials**, ensuring improved student engagement and learning outcomes.

Teacher of Record – C/C++ — August 2025 – Present
SUNY New Paltz — *New Paltz, NY*

- Assist professor in teaching foundational programming concepts using **C and C++,** supporting a class of 40+ first-year students.
- Hold weekly **office hours** to provide one-on-one academic support, clarify complex topics, and mentor students in best coding practices.
- Conduct **lab sessions** to guide students through loops, arrays, pointers, and functions while reinforcing logical problem-solving and debugging skills.
- Collaborate with faculty to design **instructional materials**, ensuring improved student engagement and learning outcomes.

Volunteer Web Developer — June 2025 – Present
Independent Project – Website Management — *New Paltz, NY*

- Maintained and optimized six live websites (two real estate, four non-profit), improving **performance, accessibility, and user experience**.
- Utilized **WordPress, PHP, MySQL, and HTML** to implement new features, fix bugs, and manage content.
- Modernized two legacy **real estate websites** (originally developed 25+ years ago) by **restructuring outdated folder hierarchies** and **updating styling/formatting** for maintainability and industry best practices.
- Configured hosting and domain services through **Network Solutions**, including **DNS and SSL** setup.
- Managed **file transfers, deployments, and backups** using **FileZilla (FTP)** and hosting control panels.
- Assisted in **streaming/media setup** and troubleshooting for client-facing websites.
- Collaborated directly with site owners to ensure smooth operation and long-term stability.

Software Engineer - Development — Dec 2023 – Feb 2024
Pansuriya Impex (HK) Ltd. — *HQ: Hong Kong*

- Contributed to the development of the company's official **enterprise web application**, supporting both **internal diamond management** operations and **external customer purchasing workflows**.
- Integrated external APIs through **RESTful services**, implementing secure authentication and data retrieval for seamless system interoperability.
- Implemented robust **data validation** and **exception handling** (try–catch blocks), persisting validated data into the database using **Java Mappers** for reliability.
- Developed a **Java Scheduler** to automate recurring method calls, ensuring uninterrupted system functionality and reducing manual intervention.
- Actively debugged and optimized existing modules, improving application **stability, performance, and user experience**.

Associate Software Engineer - Development May 2022 – Nov 2022

SAS Research & Development Pvt. Ltd. *HQ: North Carolina*

- Played a key role in developing **RESTful API endpoints** using **Java** and **Spring**, ensuring user-friendly responses with contextual links to related objects for a more intuitive user experience.
- Contributed to the **Retail Gateway** project by creating simplified **API wrappers**, improving system integration, maintainability, and performance.
- Assisted in implementing automated **unit tests** with **JUnit**, increasing code reliability and supporting regression testing for new and existing components.
- Focused on maintaining and refactoring the codebase with an emphasis on **scalability, performance, and security**, following industry best practices under senior developer mentorship.
- Collaborated in an **Agile/Scrum environment**, participating in code reviews, sprint planning, and knowledge-sharing sessions.

Computer Science Tutor Aug 2023 – Jan 2025

Self-Employed *Remote*

- Tutored ICSE (Indian Certificate of Secondary Education) students in **Core Java**, covering arrays, patterns, logic building, and **object-oriented programming (OOP)**.
- Designed structured **lesson plans** and tailored learning strategies to simplify complex concepts and improve **problem-solving skills**.
- Guided students in **debugging, code optimization, and exam preparation** through hands-on practice and real-world coding exercises.
- Provided individualized mentorship to strengthen **computational thinking** and **algorithm design fundamentals**.
- Developed custom **coding challenges and practice problems**, reinforcing concepts and fostering analytical reasoning.

PROJECTS

AI MEME GENERATOR – AI COURSE PROJECT

- Built a **Java-based** application that analyzes **natural-language** input and generates contextually relevant memes using a **weighted keyword dataset**.
- Designed an interactive **Swing UI** with GIF support, dynamic scaling, and responsive meme rendering.
- Implemented core logic for context-based meme retrieval through classes like *Meme, MemeMatcher*, and *MemeRepository*.
- Planned **grammar-based text parsing** to identify emotions and subjects for improved accuracy and humor relevance.

SOCIAL MEDIA APPLICATION – UIP COURSE PROJECT

- Developed a full-stack social media platform with **React.js** (front end), **Node.js/Express** (backend), and **MongoDB (Mongoose)**.
- Implemented secure authentication with **bcrypt.js** and **UUID**, supporting account creation, login, sessions, and protected routes.
- Designed responsive, reusable **React components** (feed, posts, interactions) with client-side routing and **RESTful APIs** for data operations.
- Utilized **dotenv** and **CORS**; streamlined development with **Nodemon** and **Concurrently**.

TO-DO APPLICATION – WDP Course Project

- Built a full-stack task management app with authentication and CRUD functionality for tasks.
- Implemented a **Node.js/Express** backend with **MySQL**, and front end using **HTML, CSS, and JavaScript**.
- Designed multiple pages (Home, About, Contact) with task creation, description editing, delete-one, and delete-all features.
- Configured with **dotenv**, enabled **CORS**, and accelerated development with **Nodemon**.

RESTAURANT MANAGEMENT SYSTEM - CDAC PROJECT

- Developed a restaurant management system using **Spring Boot**, **Spring Data JPA**, **React.js**, and **MySQL** for ordering, serving, and billing.
- Designed **RESTful APIs** and integrated **Spring Data JPA** for persistence and retrieval.
- Applied **SOLID principles** to improve scalability and maintainability.
- Conducted testing and optimization to ensure **high reliability and performance**.

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), HTML5, CSS3, C, C++, Python
Frameworks & Libraries: React.js, Node.js, Express.js, Spring Boot, Spring Data JPA, Junit, Flask, Swing
Databases: MySQL, MongoDB, PostgreSQL
Developer Tools & Platforms: Git/GitHub, Docker, FileZilla (FTP), Network Solutions (DNS/SSL/Hosting), VS Code, IntelliJ IDEA, Eclipse, Visual Studio
Other Skills: RESTful APIs, WordPress CMS, PHP, Mongoose, dotenv, CORS, bcrypt.js, UUID, Nodemon, Concurrently, NLP, Grammar-based Text Parsing